

Computer Science Questions & Answers

Five core questions with concise, accurate answers

Q1

What is the difference between a process and a thread in an operating system?

Answer:

A process is an independent program in execution with its own memory space, while a thread is a lightweight unit of execution within a process that shares the same memory and resources with other threads of the same process.

Q2

What is the purpose of a hash function in data structures and cryptography?

Answer:

A hash function maps input data of arbitrary size to a fixed-size output (hash value), enabling fast data retrieval in hash tables and ensuring data integrity and security in cryptographic applications by making it computationally infeasible to reverse or find collisions.

Q3

Explain the concept of polymorphism in Object-Oriented Programming (OOP).

Answer:

Polymorphism allows objects of different classes to be treated as objects of a common superclass, enabling a single interface to represent different underlying data types or methods — achieved through method overloading (compile-time) and method overriding (runtime).

Q4

What is the role of the TCP/IP model in computer networking?

Answer:

The TCP/IP model is a four-layer framework (Application, Transport, Internet, Network Access) that standardizes how data is transmitted over the internet by defining protocols for addressing, routing, and reliable end-to-end communication between devices.

Q5

What is the difference between supervised and unsupervised machine learning?

Answer:

Supervised learning trains a model on labeled input-output pairs to predict outcomes for new data, whereas unsupervised learning identifies hidden patterns or groupings in unlabeled data without predefined correct answers, as in clustering and dimensionality reduction.

End of Document — All answers are academically accurate and concise.